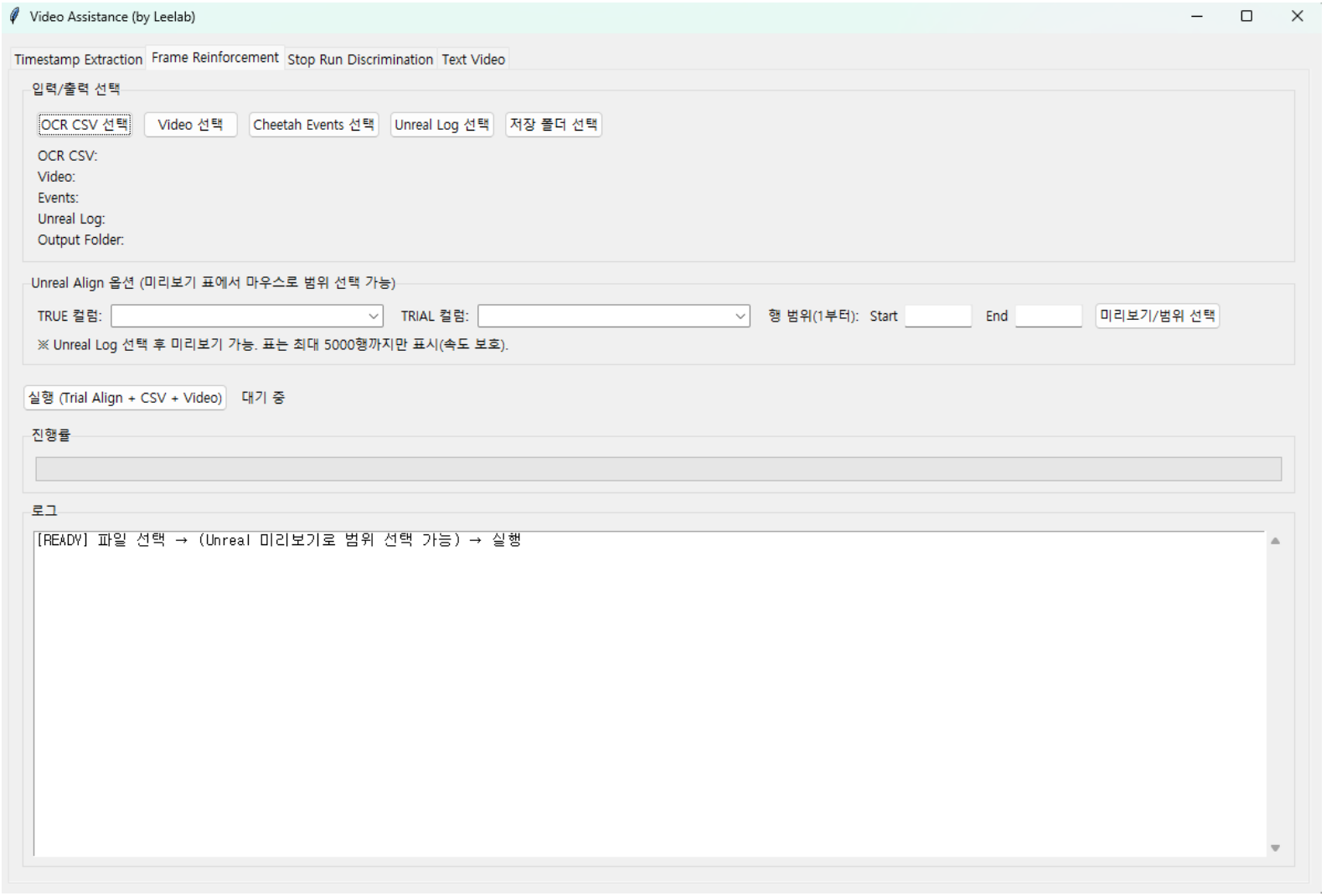
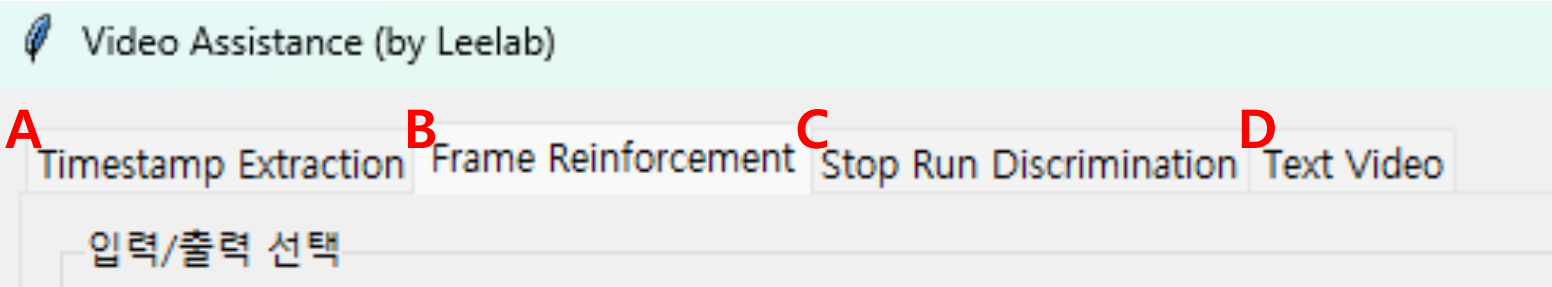
Video Assistance User Guide

Video Assistance사용



Video Assistance.py 실행

Video Assistance사용 – 기능 설명



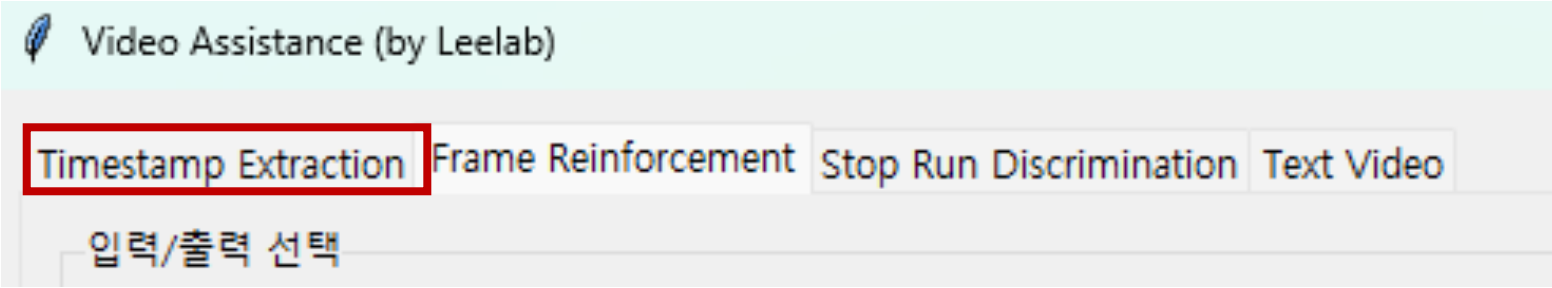
A. 영상에 출력되는 timestamp 추출

B. 영상을 unreal log trial 기준으로 frame 보강 후 보강 된 영상, CSV(Cheetah event = 영상 timestamp align) 출력

C. DLC label data를 활용하여 stop run 판별

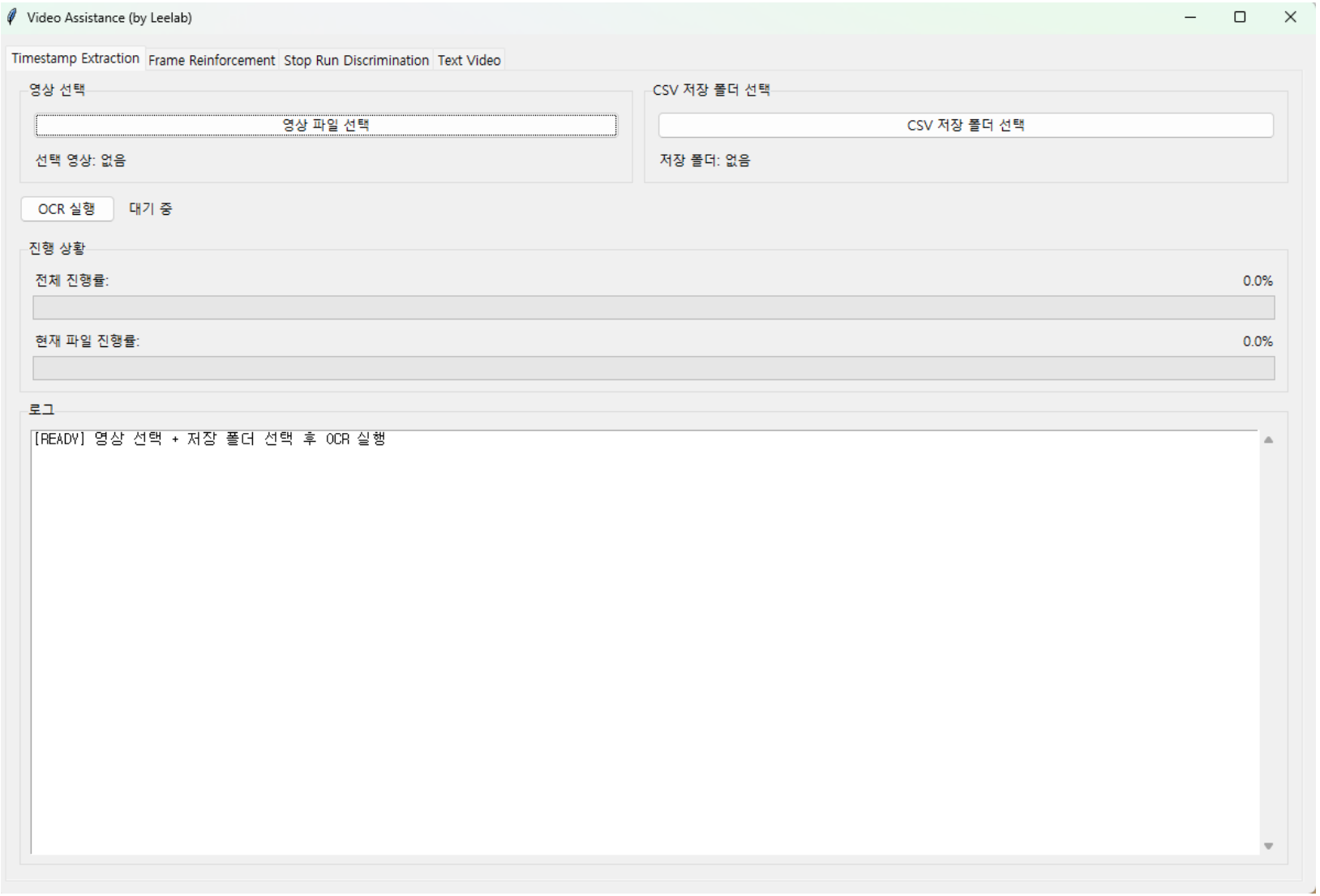
D. align 된 data를 영상과 함께 text로 출력

1. Timestamp Extraction

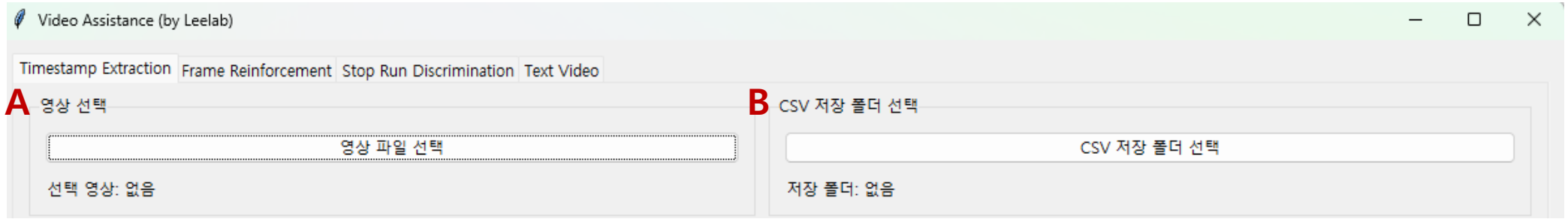


1. Timestamp Extraction

※ Timestamp Extraction 사용하기 전 사용자의 컴퓨터 GPU에 맞는 CUDA를 설치 후 사용해야 한다.



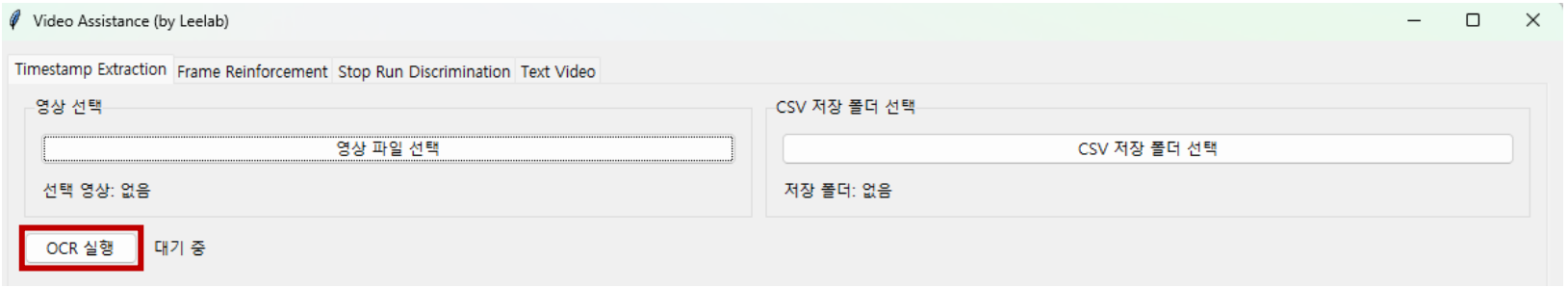
1. Timestamp Extraction



A. Timestamp를 추출할 영상 input

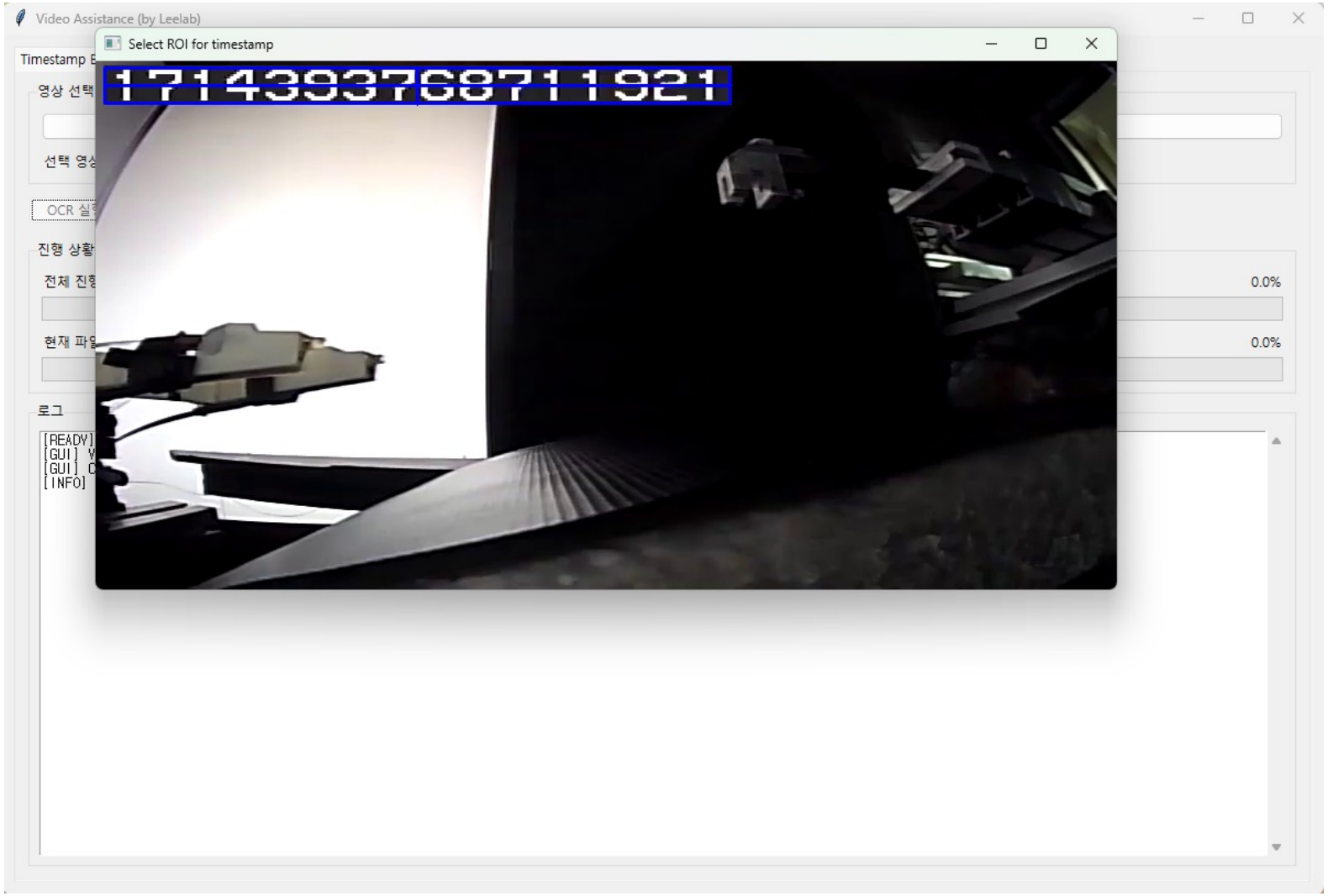
B. 추출된 timestamp csv file 저장 폴더 선택

1. Timestamp Extraction



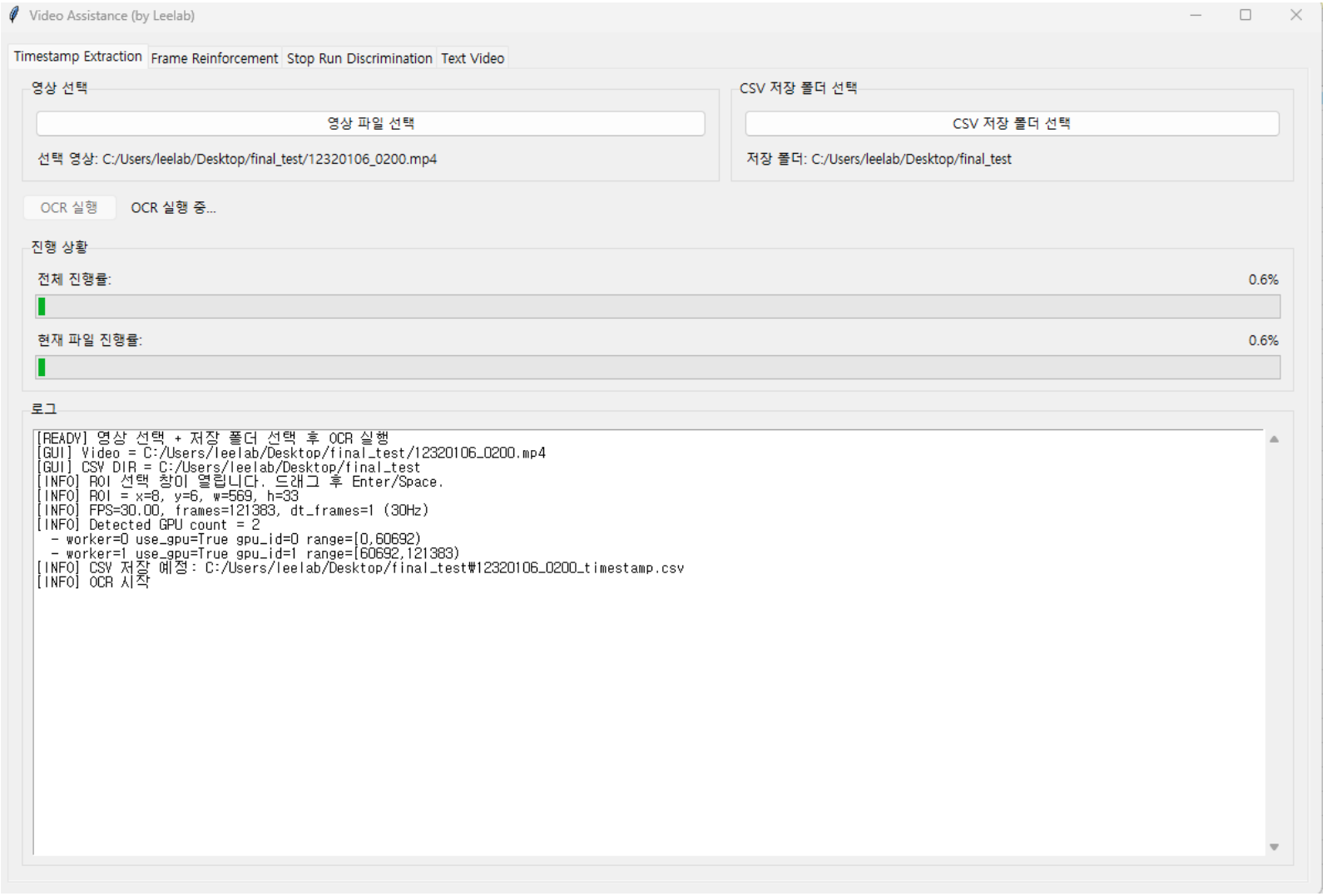
timestamp 추출영상 파일 선택, CSV 저장 폴더 선택 후 OCR 실행

1. Timestamp Extraction



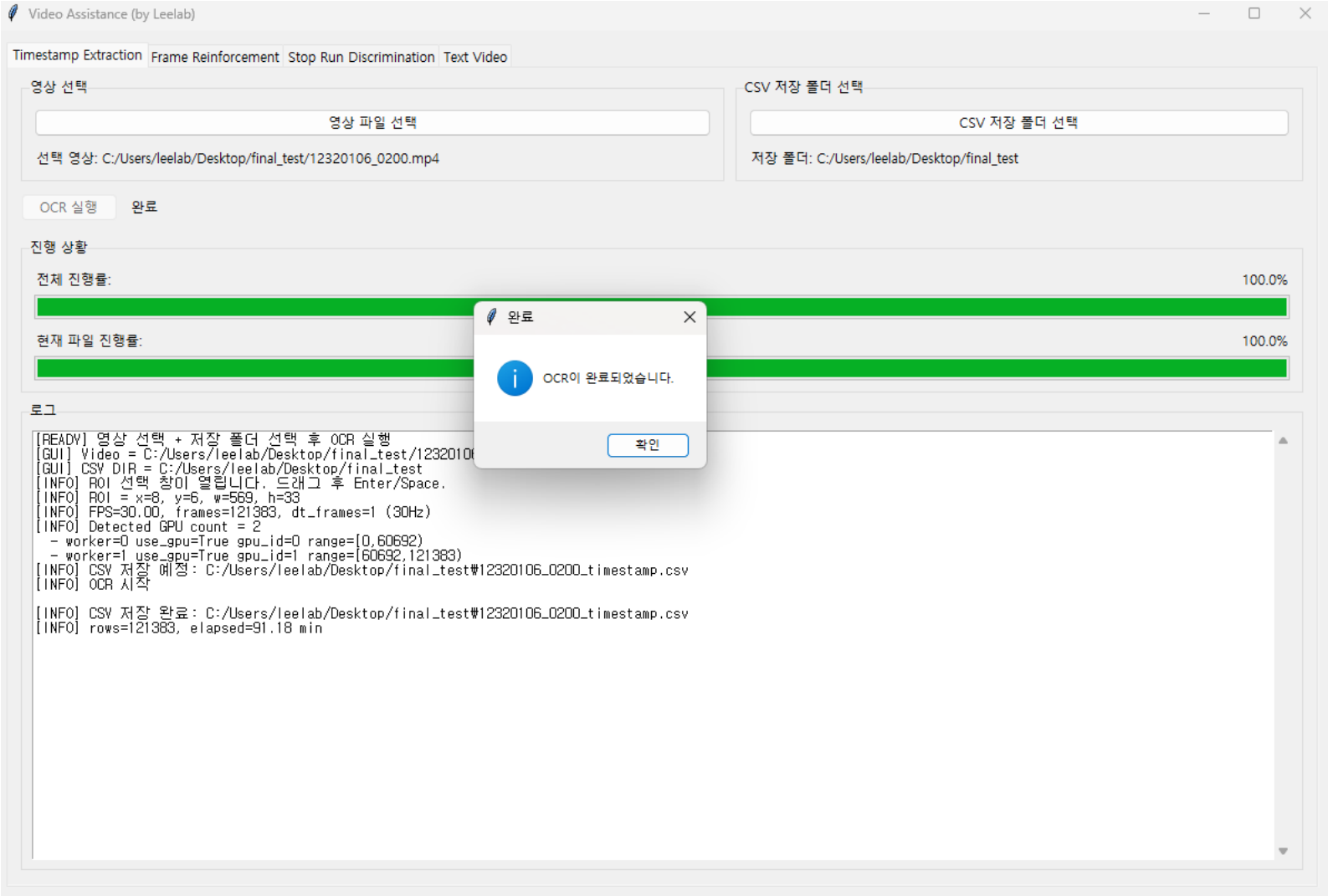
출력한 timestamp 영역을 드래그 후 enter

1. Timestamp Extraction



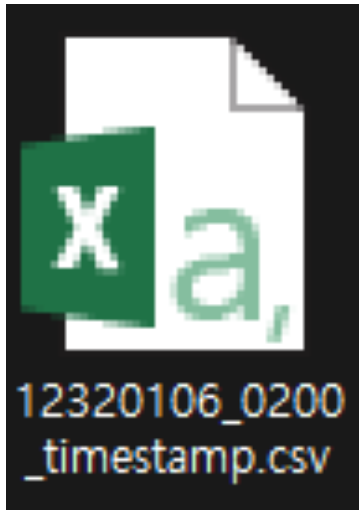
추출 진행 중

1. Timestamp Extraction



추출 완료

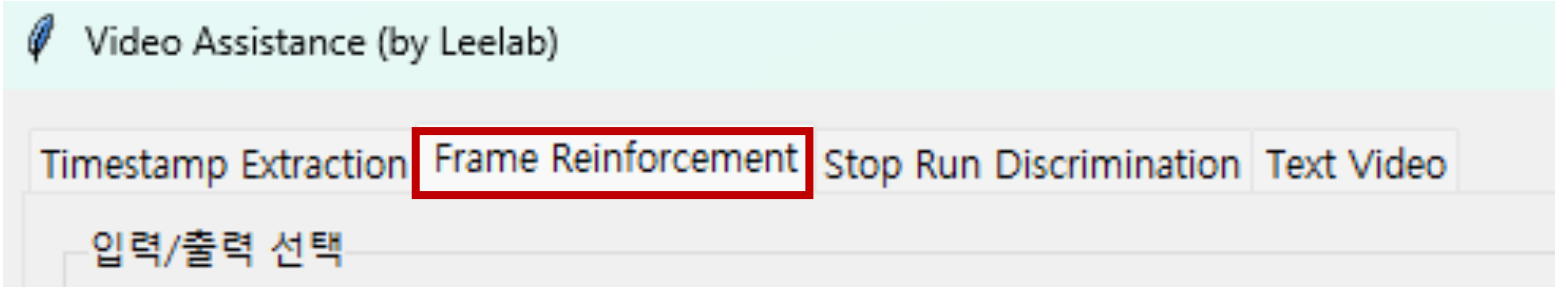
1. Timestamp Extraction



	A	B	C	D	E	F	G
1	index	time_sec	timestamp	conf	worker	used_gpu	gpu_id
2	0	0	1.71E+15	0.9	0	TRUE	0
3	1	0.033333	1.71E+15	0.9	0	TRUE	0
4	2	0.066667	1.71E+15	0.9	0	TRUE	0
5	3	0.1	1.71E+15	0.9	0	TRUE	0
6	4	0.133333	1.71E+15	0.9	0	TRUE	0
7	5	0.166667	1.71E+15	0.9	0	TRUE	0
8	6	0.2	1.71E+15	0.9	0	TRUE	0
9	7	0.233333	1.71E+15	0.9	0	TRUE	0
10	8	0.266667	1.71E+15	0.9	0	TRUE	0
11	9	0.3	1.71E+15	0.9	0	TRUE	0
12	10	0.333333	1.71E+15	0.9	0	TRUE	0
13	11	0.366667	1.71E+15	0.9	0	TRUE	0
14	12	0.4	1.71E+15	0.9	0	TRUE	0
15	13	0.433333	1.71E+15	0.9	0	TRUE	0
16	14	0.466667	1.71E+15	0.9	0	TRUE	0
17	15	0.5	1.71E+15	0.9	0	TRUE	0
18	16	0.533333	1.71E+15	0.9	0	TRUE	0
19	17	0.566667	1.71E+15	0.9	0	TRUE	0
20	18	0.6	1.71E+15	0.9	0	TRUE	0
21	19	0.633333	1.71E+15	0.9	0	TRUE	0
22	20	0.666667	1.71E+15	0.9	0	TRUE	0
23	21	0.7	1.71E+15	0.9	0	TRUE	0
24	22	0.733333	1.71E+15	0.9	0	TRUE	0
25	23	0.766667	1.71E+15	0.9	0	TRUE	0
26	24	0.8	1.71E+15	0.9	0	TRUE	0
27	25	0.833333	1.71E+15	0.9	0	TRUE	0
28	26	0.866667	1.71E+15	0.9	0	TRUE	0
29	27	0.9	1.71E+15	0.9	0	TRUE	0
30	28	0.933333	1.71E+15	0.9	0	TRUE	0
31	29	0.966667	1.71E+15	0.9	0	TRUE	0

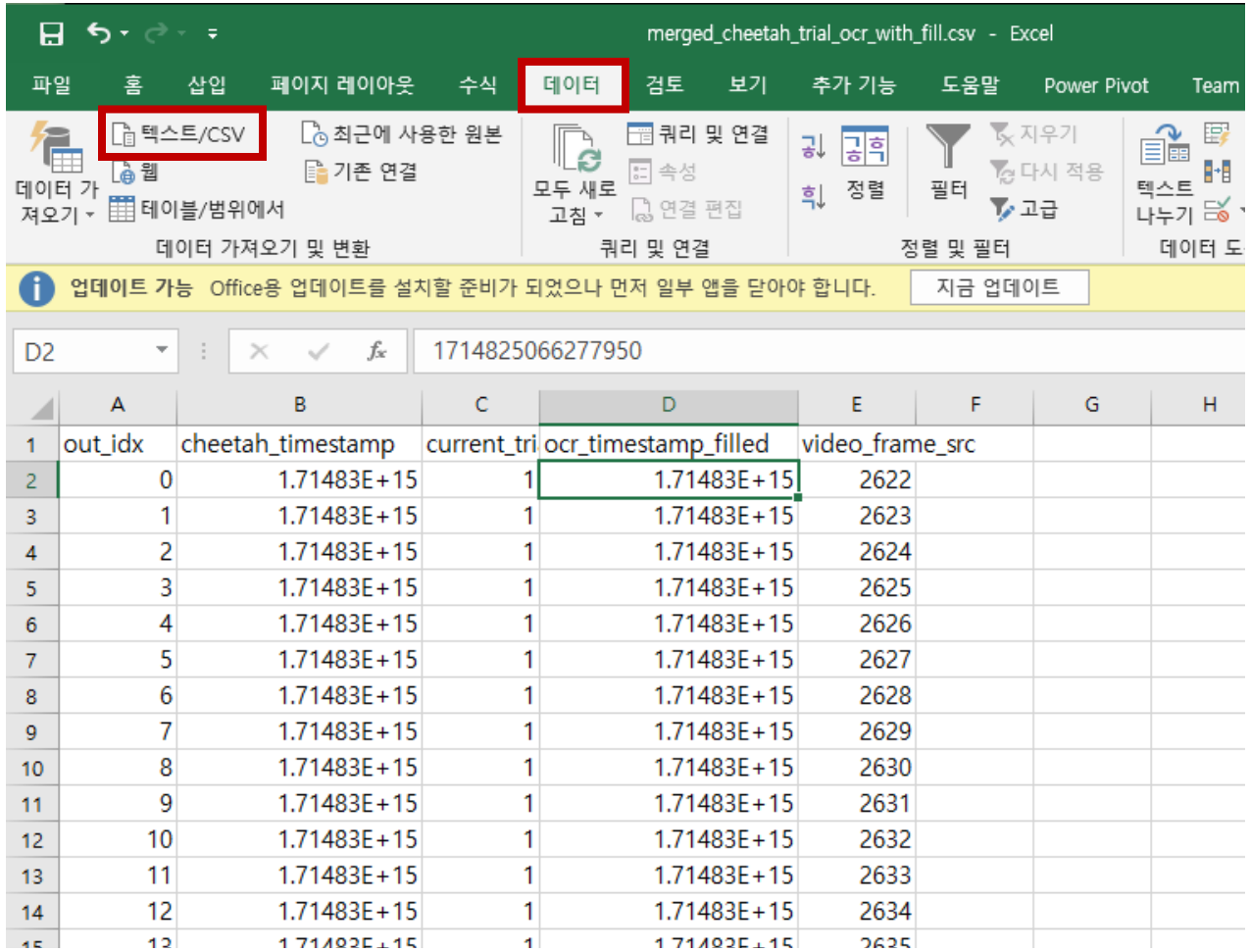
추출된 timestamp csv file

2. Frame Reinforcement



2. Frame Reinforcement – 확장자 변환

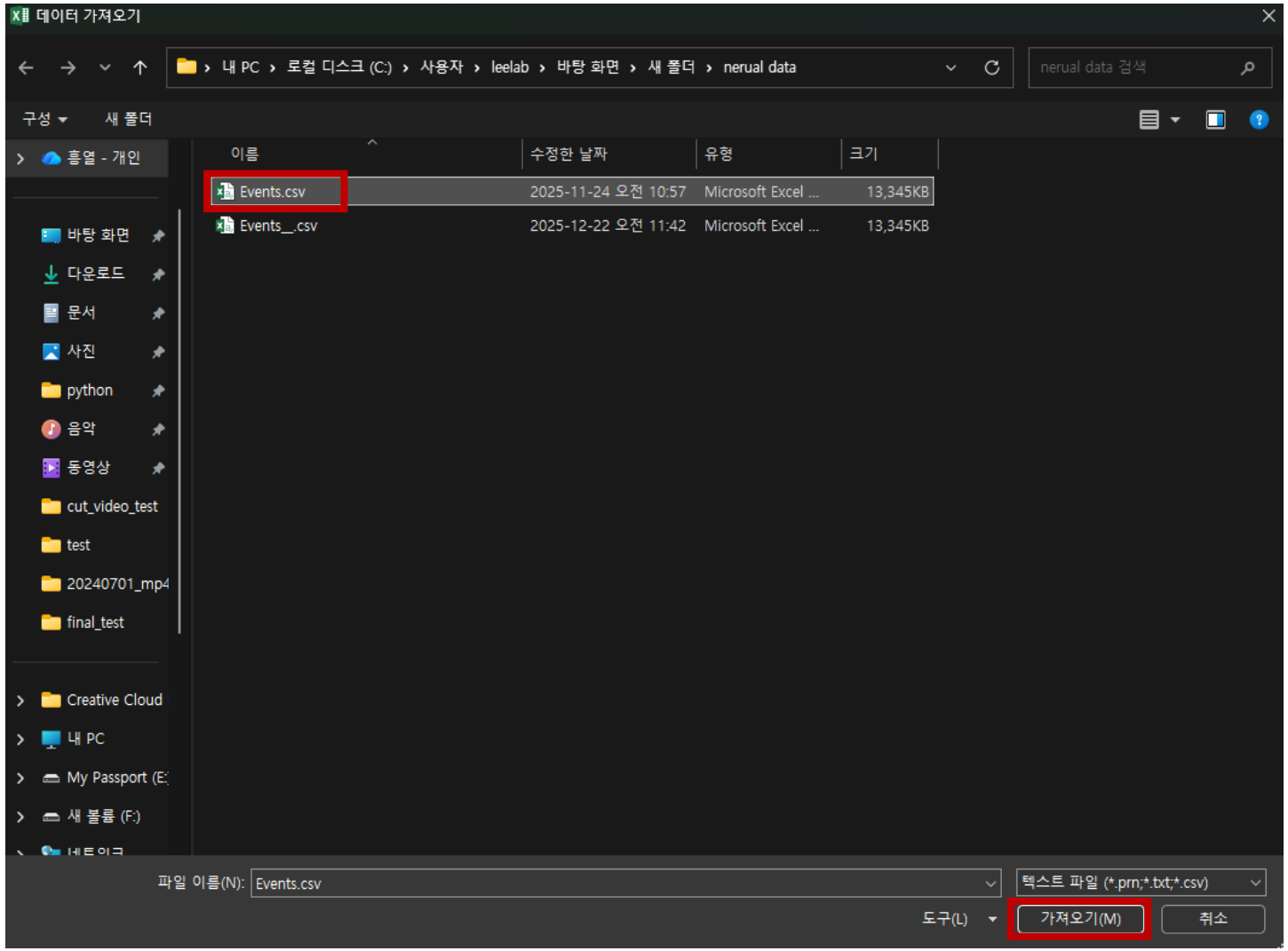
※ Frame Reinforcement 사용하기 전 cheetah event log의 csv file을 xlsx 형식으로 변환을 진행해야 한다.



	A	B	C	D	E
1	out_idx	cheetah_timestamp	current_tri	ocr_timestamp_filled	video_frame_src
2	0	1.71483E+15	1	1.71483E+15	2622
3	1	1.71483E+15	1	1.71483E+15	2623
4	2	1.71483E+15	1	1.71483E+15	2624
5	3	1.71483E+15	1	1.71483E+15	2625
6	4	1.71483E+15	1	1.71483E+15	2626
7	5	1.71483E+15	1	1.71483E+15	2627
8	6	1.71483E+15	1	1.71483E+15	2628
9	7	1.71483E+15	1	1.71483E+15	2629
10	8	1.71483E+15	1	1.71483E+15	2630
11	9	1.71483E+15	1	1.71483E+15	2631
12	10	1.71483E+15	1	1.71483E+15	2632
13	11	1.71483E+15	1	1.71483E+15	2633
14	12	1.71483E+15	1	1.71483E+15	2634

Cheetah event log.csv에서 데이터 – 텍스트/CSV 선택

2. Frame Reinforcement – 확장자 변환



cheetah event log.csv 가져오기 선택

2. Frame Reinforcement – 확장자 변환

Events.csv

파일 원본: 949: 한국어

구분 기호: 쉼표

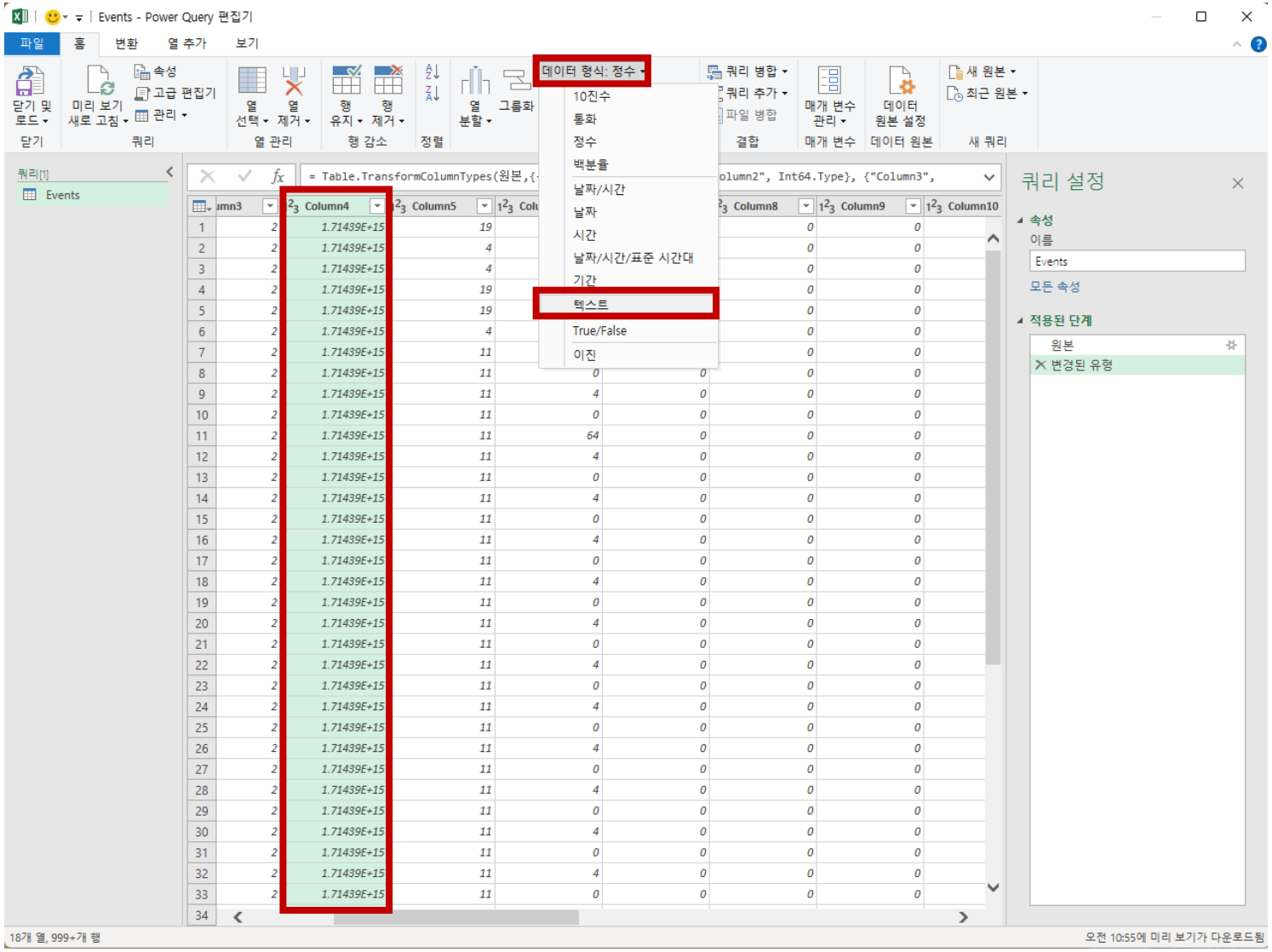
데이터 형식 검색: 첫 200개 행 기준

Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9	Column10	Column11	Co
0	0	2	1.71439E+15	19	0	0	0	0	0	0	
0	0	2	1.71439E+15	4	0	0	0	0	0	0	
0	0	2	1.71439E+15	4	0	0	0	0	0	0	
0	0	2	1.71439E+15	19	0	0	0	0	0	0	
0	0	2	1.71439E+15	19	0	0	0	0	0	0	
0	0	2	1.71439E+15	4	0	0	0	0	0	0	
0	0	2	1.71439E+15	11	4	0	0	0	0	0	
0	0	2	1.71439E+15	11	0	0	0	0	0	0	
0	0	2	1.71439E+15	11	4	0	0	0	0	0	
0	0	2	1.71439E+15	11	0	0	0	0	0	0	
0	0	2	1.71439E+15	11	64	0	0	0	0	0	
0	0	2	1.71439E+15	11	4	0	0	0	0	0	
0	0	2	1.71439E+15	11	0	0	0	0	0	0	
0	0	2	1.71439E+15	11	4	0	0	0	0	0	
0	0	2	1.71439E+15	11	0	0	0	0	0	0	
0	0	2	1.71439E+15	11	4	0	0	0	0	0	
0	0	2	1.71439E+15	11	0	0	0	0	0	0	
0	0	2	1.71439E+15	11	4	0	0	0	0	0	
0	0	2	1.71439E+15	11	0	0	0	0	0	0	
0	0	2	1.71439E+15	11	4	0	0	0	0	0	

로드 | 데이터 변환 | 취소

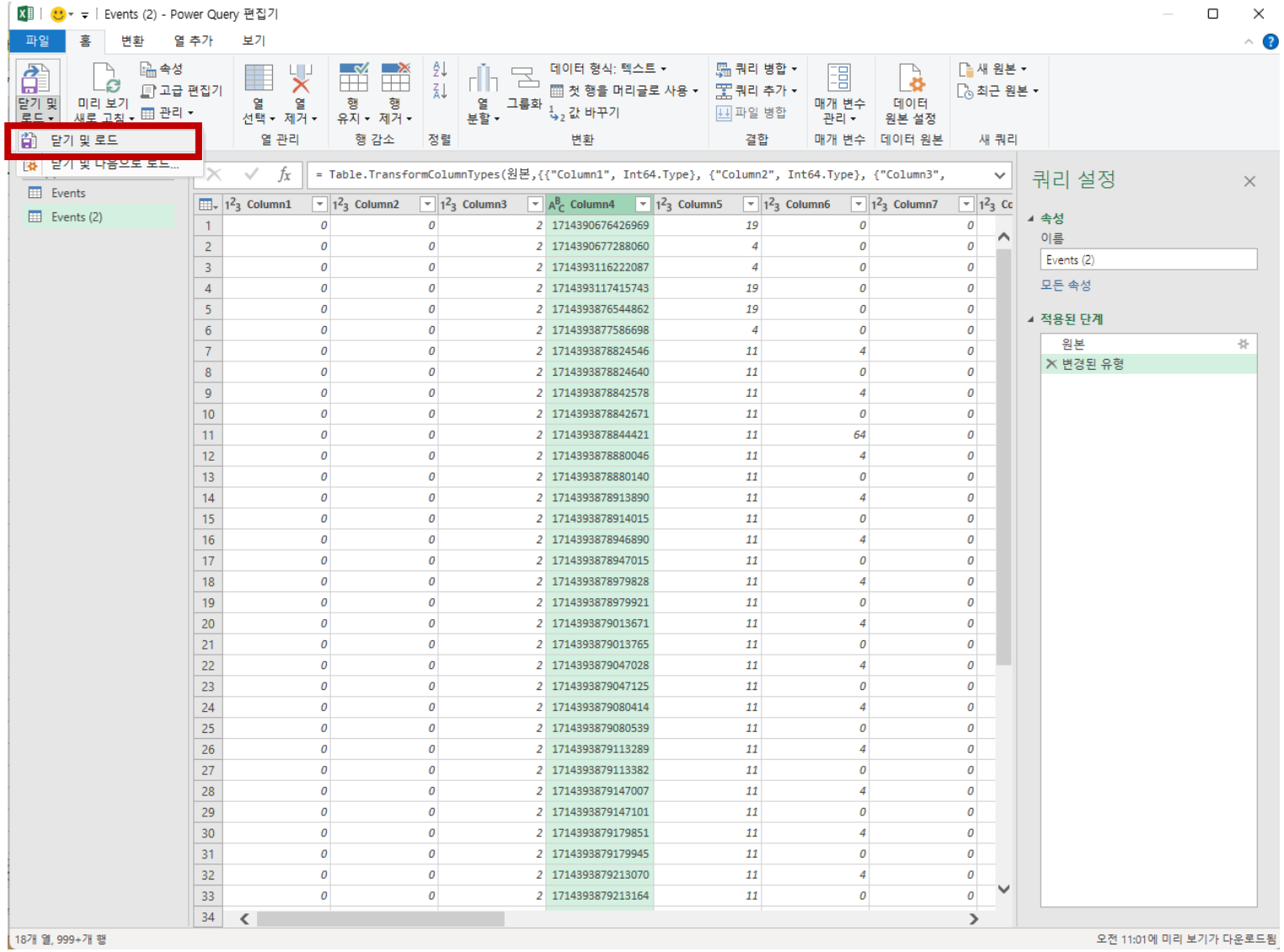
데이터 변환 선택

2. Frame Reinforcement – 확장자 변환



timestamp 선택 후 데이터 형식 – 텍스트 선택

2. Frame Reinforcement – 확장자 변환



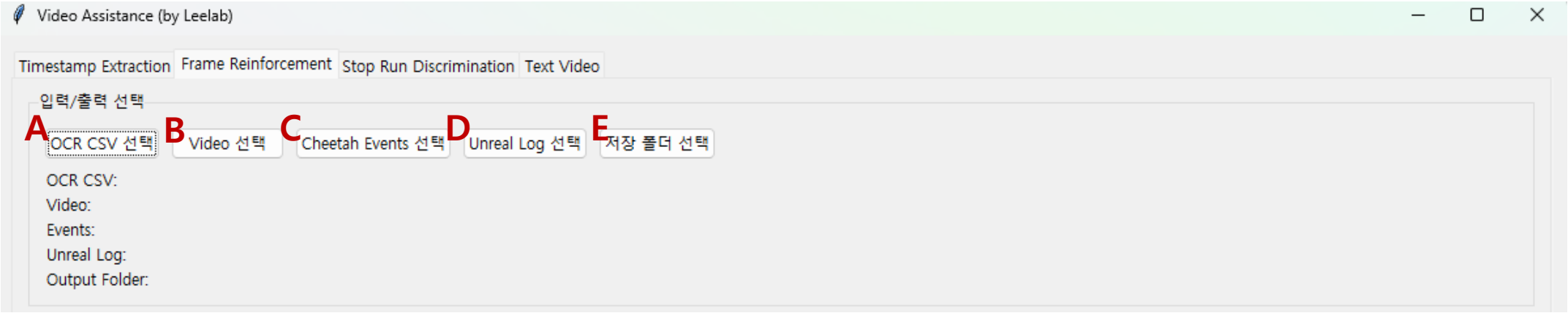
닫기 및 로드

2. Frame Reinforcement – 확장자 변환

Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9	Column10	Column11
0	0	2	1714390676426969	19	0	0	0	0	0	0
0	0	2	1714390677288060	4	0	0	0	0	0	0
0	0	2	1714393116222087	4	0	0	0	0	0	0
0	0	2	1714393117415743	19	0	0	0	0	0	0
0	0	2	1714393876544862	19	0	0	0	0	0	0
0	0	2	1714393877586698	4	0	0	0	0	0	0
0	0	2	1714393878824546	11	4	0	0	0	0	0
0	0	2	1714393878824640	11	0	0	0	0	0	0
0	0	2	1714393878842578	11	4	0	0	0	0	0
0	0	2	1714393878842671	11	0	0	0	0	0	0
0	0	2	1714393878844421	11	64	0	0	0	0	0
0	0	2	1714393878880046	11	4	0	0	0	0	0
0	0	2	1714393878880140	11	0	0	0	0	0	0
0	0	2	1714393878913890	11	4	0	0	0	0	0
0	0	2	1714393878914015	11	0	0	0	0	0	0
0	0	2	1714393878946890	11	4	0	0	0	0	0
0	0	2	1714393878947015	11	0	0	0	0	0	0
0	0	2	1714393878979828	11	4	0	0	0	0	0
0	0	2	1714393878979921	11	0	0	0	0	0	0
0	0	2	1714393879013671	11	4	0	0	0	0	0
0	0	2	1714393879013765	11	0	0	0	0	0	0
0	0	2	1714393879047028	11	4	0	0	0	0	0
0	0	2	1714393879047125	11	0	0	0	0	0	0
0	0	2	1714393879080414	11	4	0	0	0	0	0
0	0	2	1714393879080539	11	0	0	0	0	0	0
0	0	2	1714393879113289	11	4	0	0	0	0	0
0	0	2	1714393879113382	11	0	0	0	0	0	0
0	0	2	1714393879147007	11	4	0	0	0	0	0
0	0	2	1714393879147101	11	0	0	0	0	0	0
0	0	2	1714393879179851	11	4	0	0	0	0	0
0	0	2	1714393879179945	11	0	0	0	0	0	0
0	0	2	1714393879213070	11	4	0	0	0	0	0

원본의 cheetah event log.csv를 삭제 후 다른 이름 저장(.xlsx 확장자 저장)

2. Frame Reinforcement



A. Timestamp Extraction에서 출력된 timestamp csv file input

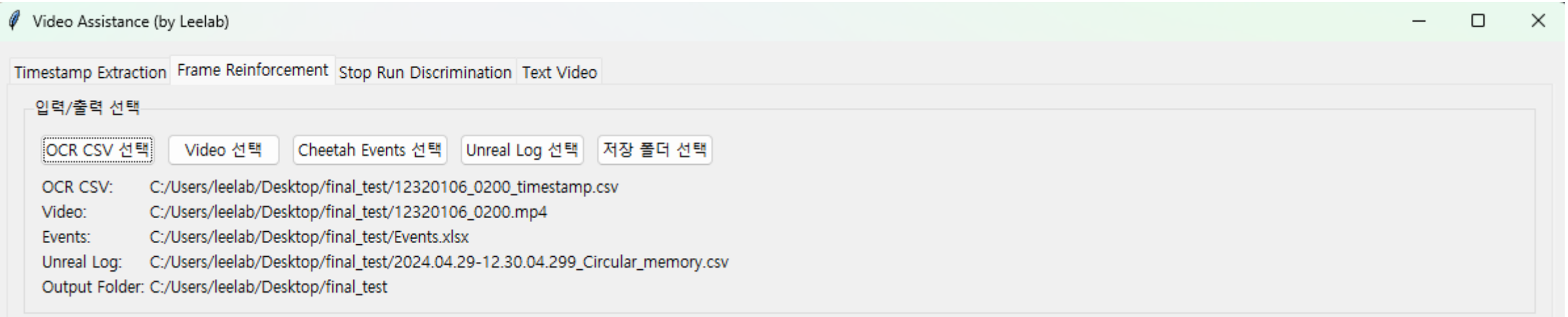
B. trial 기준으로 frame을 보강 할 영상 input

C. Cheetah event log input **(.xlsx 확장자 변환 후 사용할 것)**

D. unreal log input

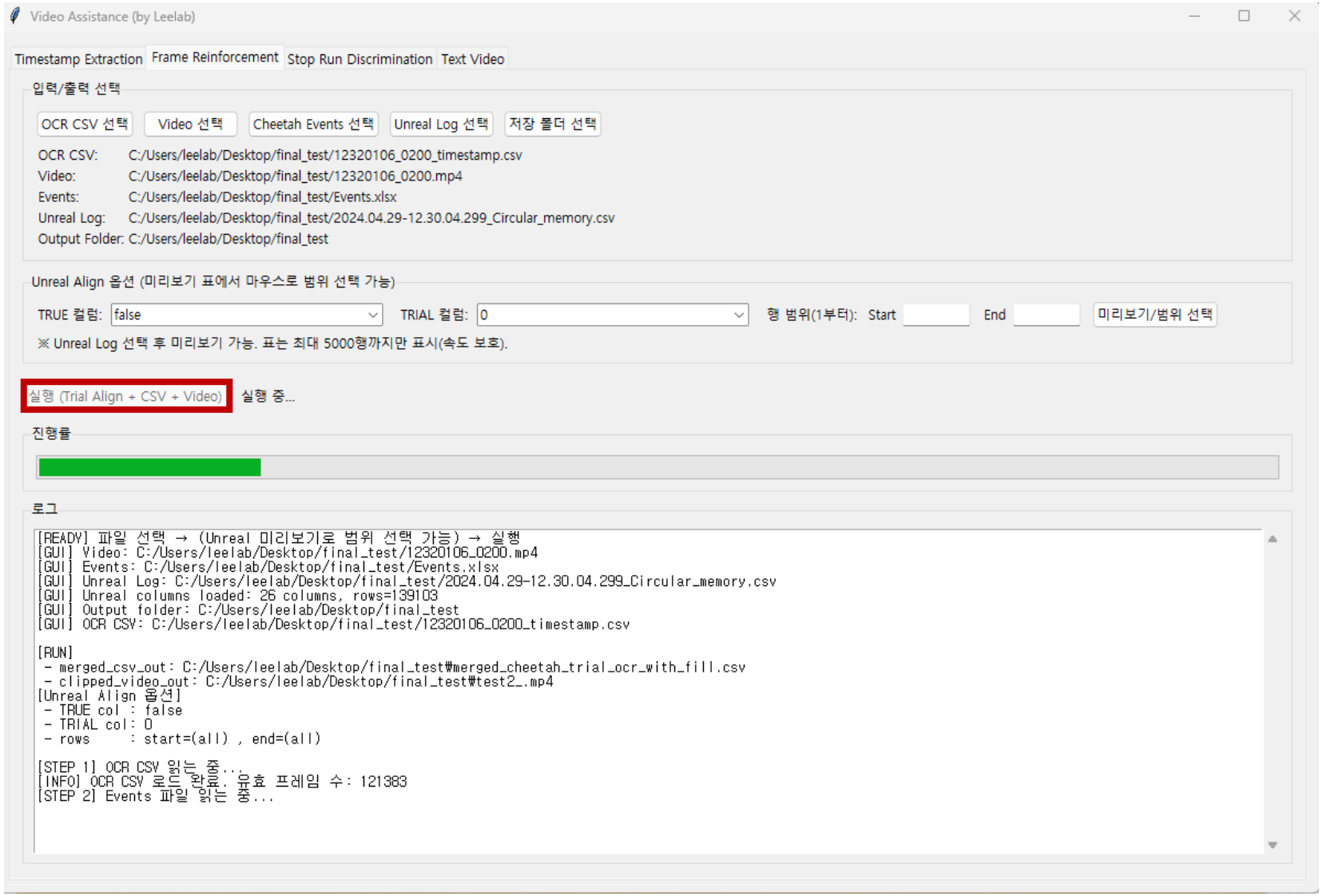
E. output(frame이 보강된 영상, trial timestamp가 align 된 csv file) 저장 경로 선택

2. Frame Reinforcement



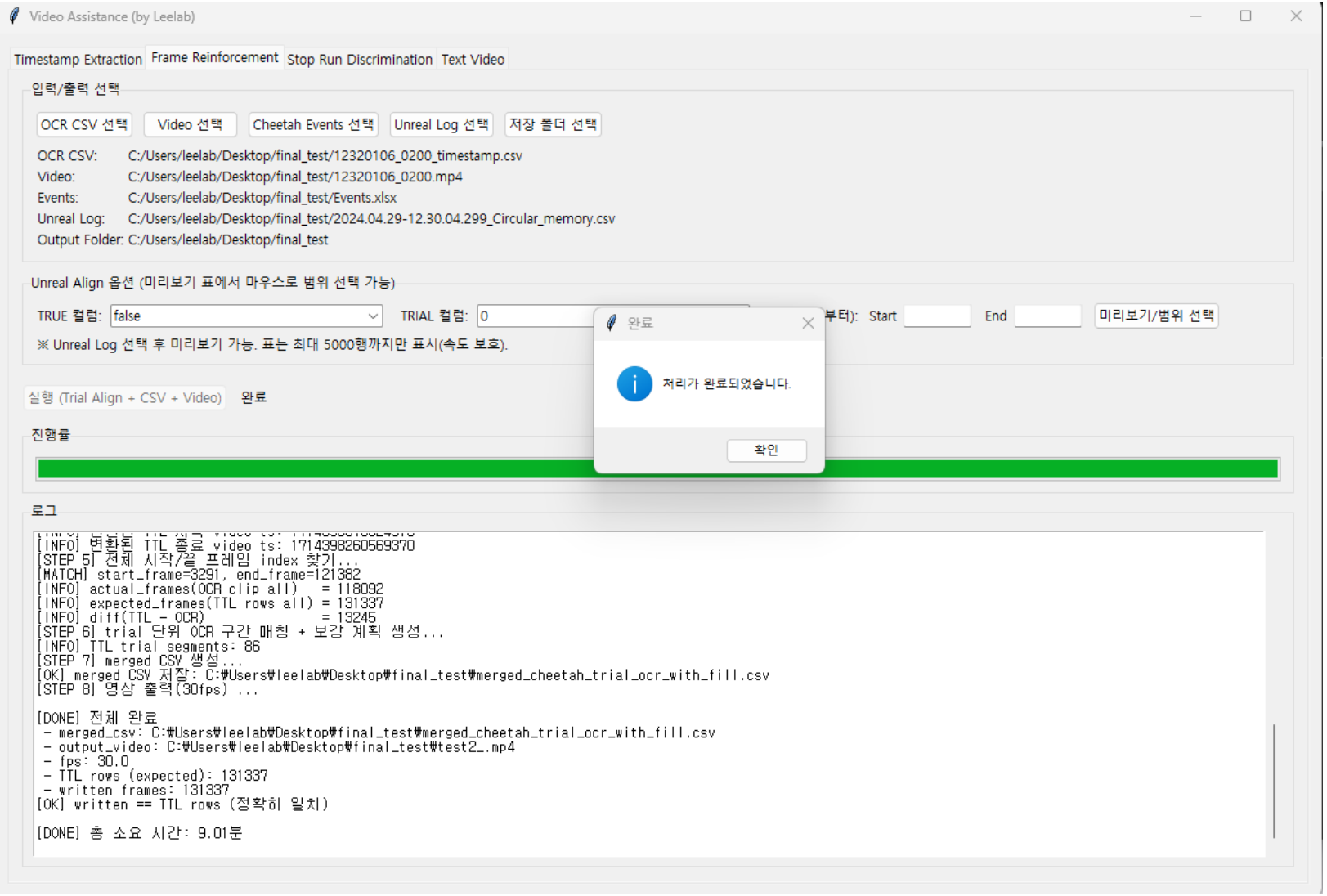
input data 경로 확인

2. Frame Reinforcement



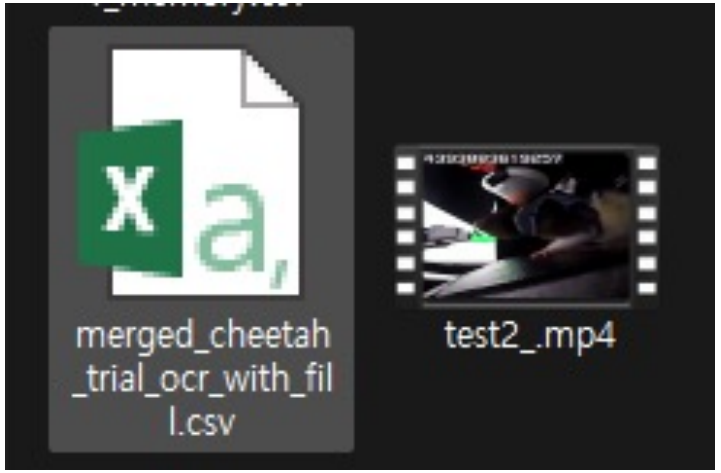
실행

2. Frame Reinforcement



실행 완료

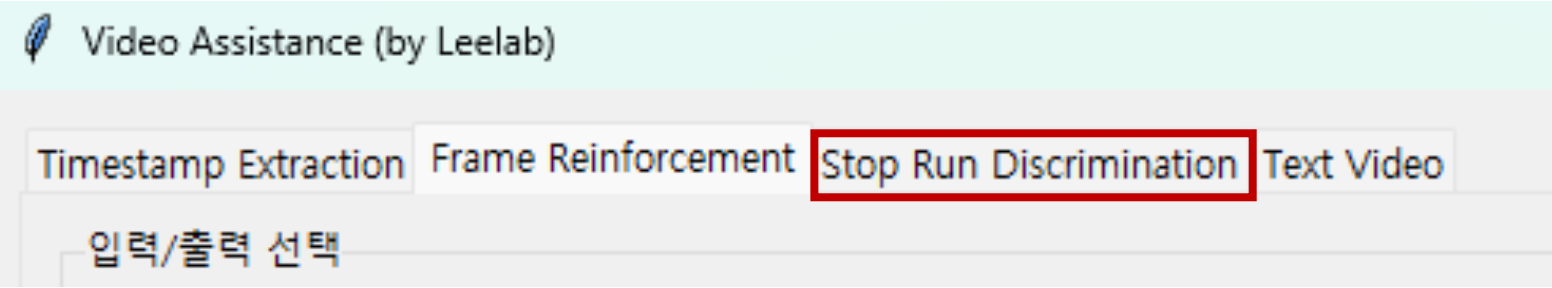
2. Frame Reinforcement



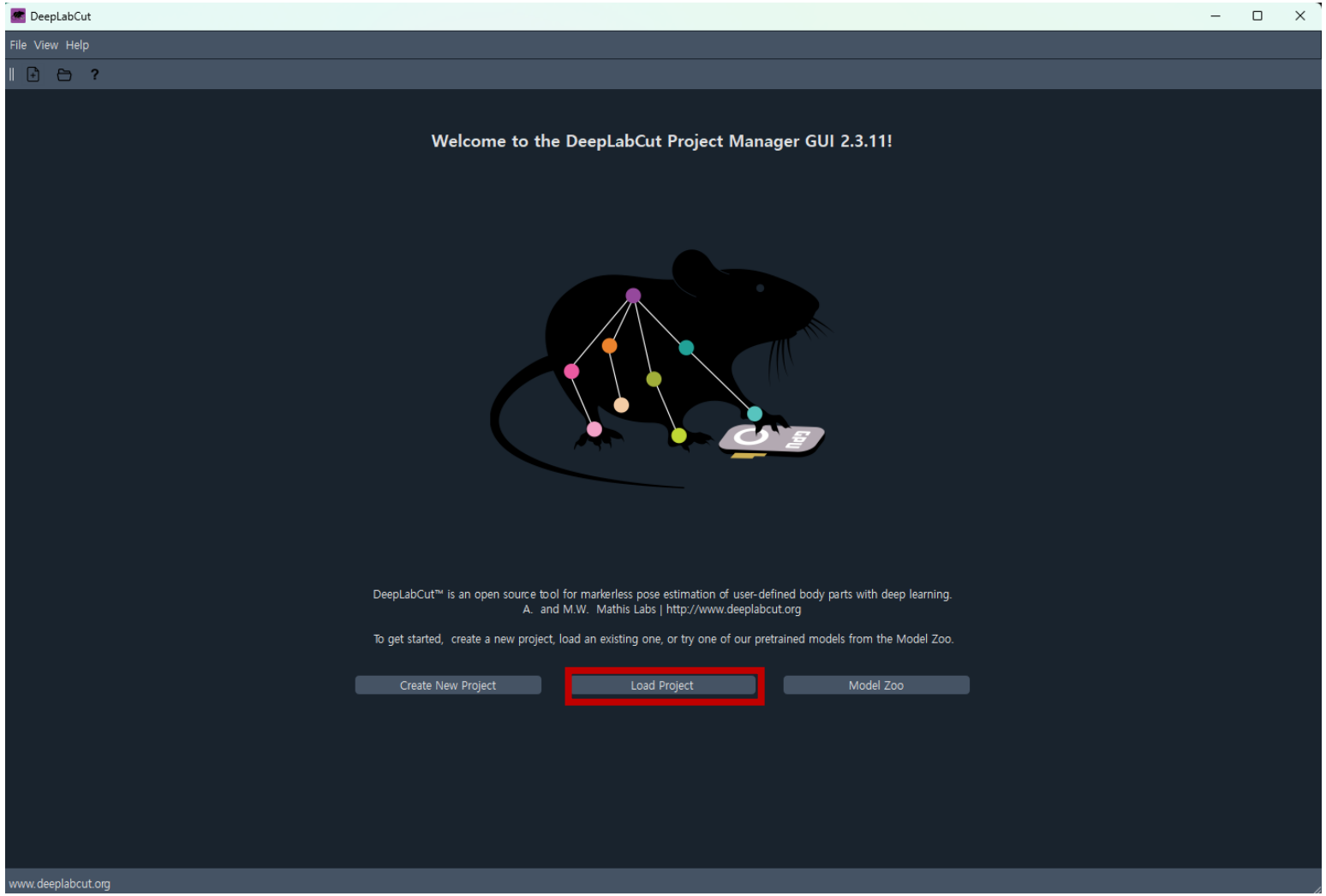
	A	B	C	D	E	F
1	out_idx	cheetah_ti	current_trial	ocr_timestamp_filled	video_frame_src	
2	0	1.71E+15	1	1.71439E+15	3291	
3	1	1.71E+15	1	1.71439E+15	3292	
4	2	1.71E+15	1	1.71439E+15	3293	
5	3	1.71E+15	1	1.71439E+15	3294	
6	4	1.71E+15	1	1.71439E+15	3295	
7	5	1.71E+15	1	1.71439E+15	3296	
8	6	1.71E+15	1	1.71439E+15	3297	
9	7	1.71E+15	1	1.71439E+15	3298	
10	8	1.71E+15	1	1.71439E+15	3299	
11	9	1.71E+15	1	1.71439E+15	3300	
12	10	1.71E+15	1	1.71439E+15	3301	
13	11	1.71E+15	1	1.71439E+15	3302	
14	12	1.71E+15	1	1.71439E+15	3303	
15	13	1.71E+15	1	1.71439E+15	3304	
16	14	1.71E+15	1	1.71439E+15	3305	
17	15	1.71E+15	1	1.71439E+15	3306	
18	16	1.71E+15	1	1.71439E+15	3307	
19	17	1.71E+15	1	1.71439E+15	3308	
20	18	1.71E+15	1	1.71439E+15	3309	
21	19	1.71E+15	1	1.71439E+15	3310	
22	20	1.71E+15	1	1.71439E+15	3311	
23	21	1.71E+15	1	1.71439E+15	3312	
24	22	1.71E+15	1	1.71439E+15	3313	
25	23	1.71E+15	1	1.71439E+15	3314	
26	24	1.71E+15	1	1.71439E+15	3315	
27	25	1.71E+15	1	1.71439E+15	3316	
28	26	1.71E+15	1	1.71439E+15	3317	

Frame이 보강된 영상과 align 정보가 적힌 csv file 출력

3. Stop Run Discrimination

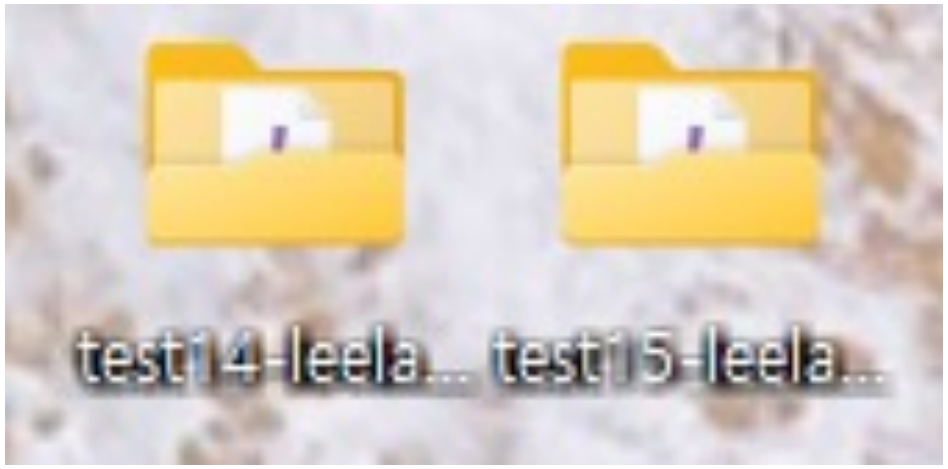


3. Stop Run Discrimination - DLC



DLC를 실행시킨 후 Load project

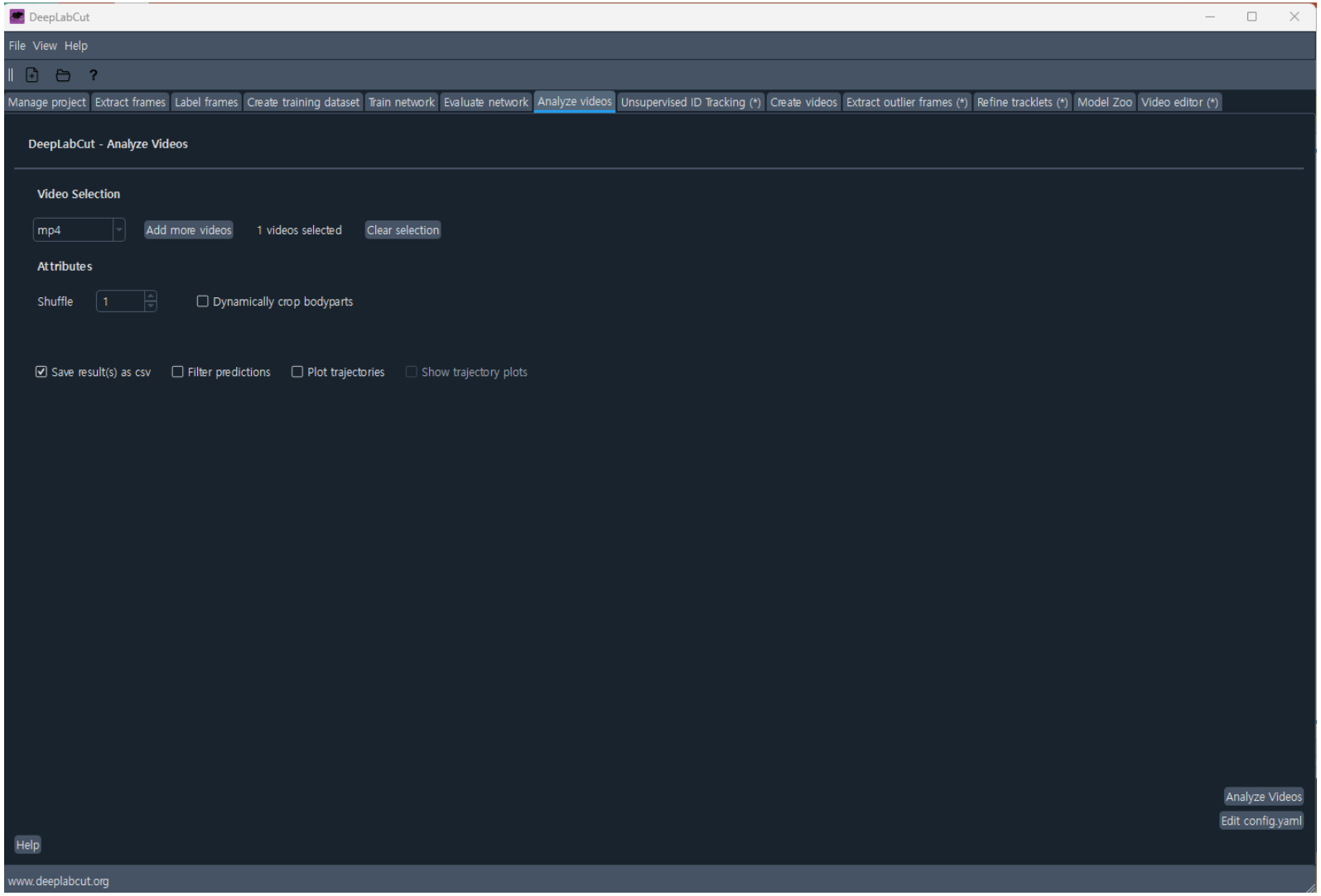
3. Stop Run Discrimination - DLC



test14 – 측면 영상 project
test15 – 정면 영상 project

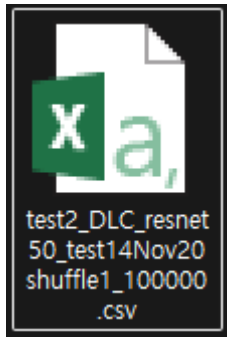
분석 영상에 맞는 project 불러오기
※불러오는 방법은 DLC사용가이드 참고

3. Stop Run Discrimination - DLC



DLC analyze videos를 통해 Frame Reinforcement로 출력된 영상을 넣어 DLC label data 출력

3. Stop Run Discrimination - DLC



	A	B	C	D	E	F	G	H	I	J
1	scorer	DLC_resne	DLC_resne	DLC_resne	DLC_resne	DLC_resne	DLC_resnet50_test14Nov20shuffle1_100000			
2	bodyparts	left_leg	left_leg	left_leg	right_leg	right_leg	right_leg			
3	coords	x	y	likelihood	x	y	likelihood			
4	0	580.5129	322.6625	0.854215	578.6444	323.3225	0.87977			
5	1	482.0904	300.2362	0.998382	493.0425	295.6533	0.010099			
6	2	385.2881	326.6183	0.974744	487.0127	325.5013	0.151552			
7	3	473.9662	345.544	0.999179	476.1042	340.2759	0.013264			
8	4	608.0491	322.0126	0.989371	619.1069	318.4738	0.044213			
9	5	499.4873	298.8803	0.989501	721.2505	118.8361	0.014961			
10	6	415.8175	315.5934	0.926241	417.9351	311.8076	0.94852			
11	7	456.3912	341.0372	0.107016	458.2773	338.6851	0.96814			
12	8	561.1155	325.9793	0.957214	563.1508	324.2196	0.701022			
13	9	494.8902	294.257	0.99909	719.1444	120.099	0.007562			
14	10	393.5537	317.7222	0.438722	501.605	318.6631	0.4843			
15	11	473.399	345.2799	0.998295	479.6003	341.554	0.02046			
16	12	601.7471	326.5215	0.983877	607.8585	324.1533	0.028957			
17	13	508.1038	303.5594	0.990144	514.5222	300.2678	0.041966			
18	14	418.0007	312.5953	0.734158	418.3831	311.6598	0.989552			
19	15	449.9694	343.4333	0.958487	451.8745	341.2004	0.034408			
20	16	553.4557	328.569	0.976488	559.1426	326.8525	0.495223			
21	17	501.3015	294.9591	0.999394	510.4589	293.3834	0.011949			
22	18	396.229	318.7567	0.071565	505.6002	319.8571	0.262194			
23	19	464.2526	348.5561	0.99617	467.7665	344.4148	0.033236			

영상 label 좌표 data 출력

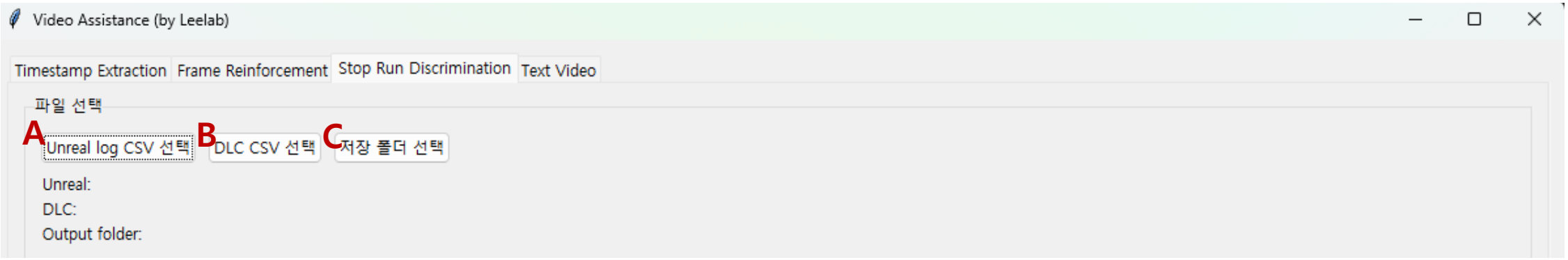
3. Stop Run Discrimination - DLC

※다른 컴퓨터에서 DLC를 사용한다면?

```
bodyparts:
- left_leg
- right_leg
```

Bodyparts의 명창만 left_leg, right_leg로 설정한 뒤 학습시킨 DLC를 사용하면 가능

3. Stop Run Discrimination

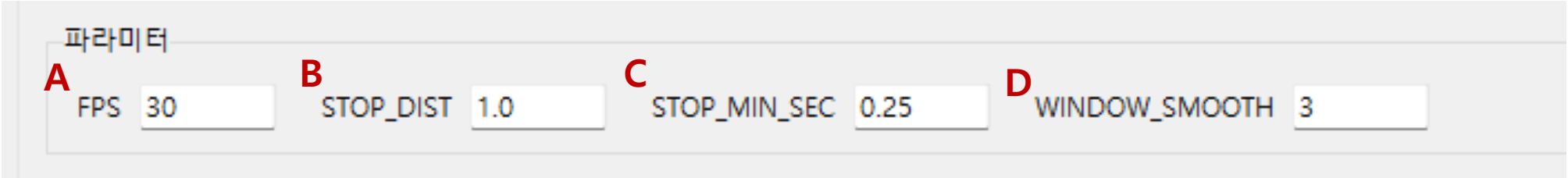


A. unreal log csv input

B. DLC로 출력된 label 좌표가 존재하는 csv input

C. 저장 경로 선택

3. Stop Run Discrimination



ex) 30fps의 영상에서 사용한 DLC 좌표 data를 가지고 중앙 좌표의 smooth를 3을 주고 1pixel 이하의 움직임이 0.25초 동안 지속될 경우 stop으로 판단한다

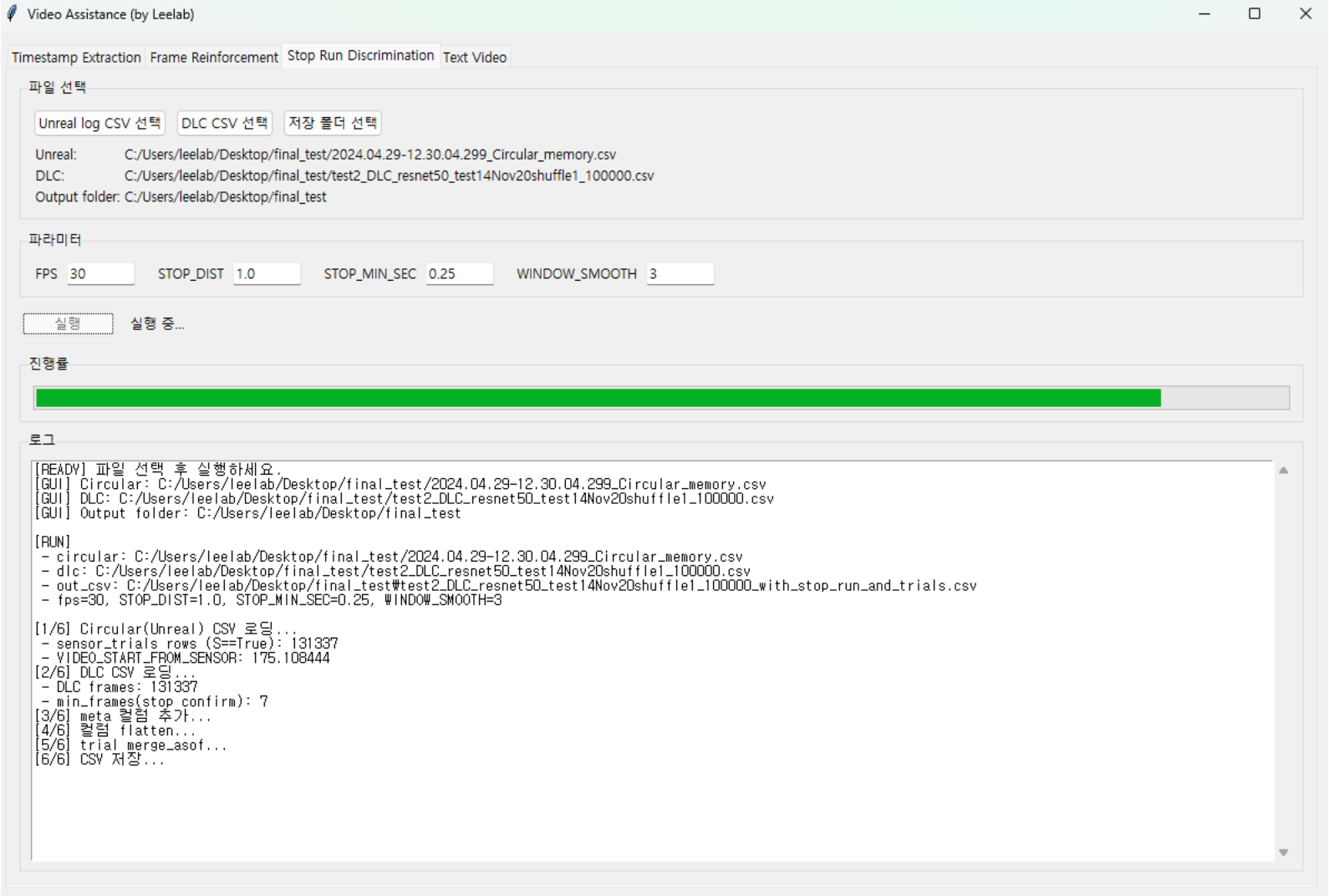
A. FPS 30으로 고정 – 영상의 fps에 따라 설정 (30이 기본 조건)

B. label jitter의 허용 범위 (1.0이 기본 조건)

C. stop 조건 – Jitter 허용 범위가 C초 이상 지속될 경우 stop으로 판별 (0.25가 기본 조건)

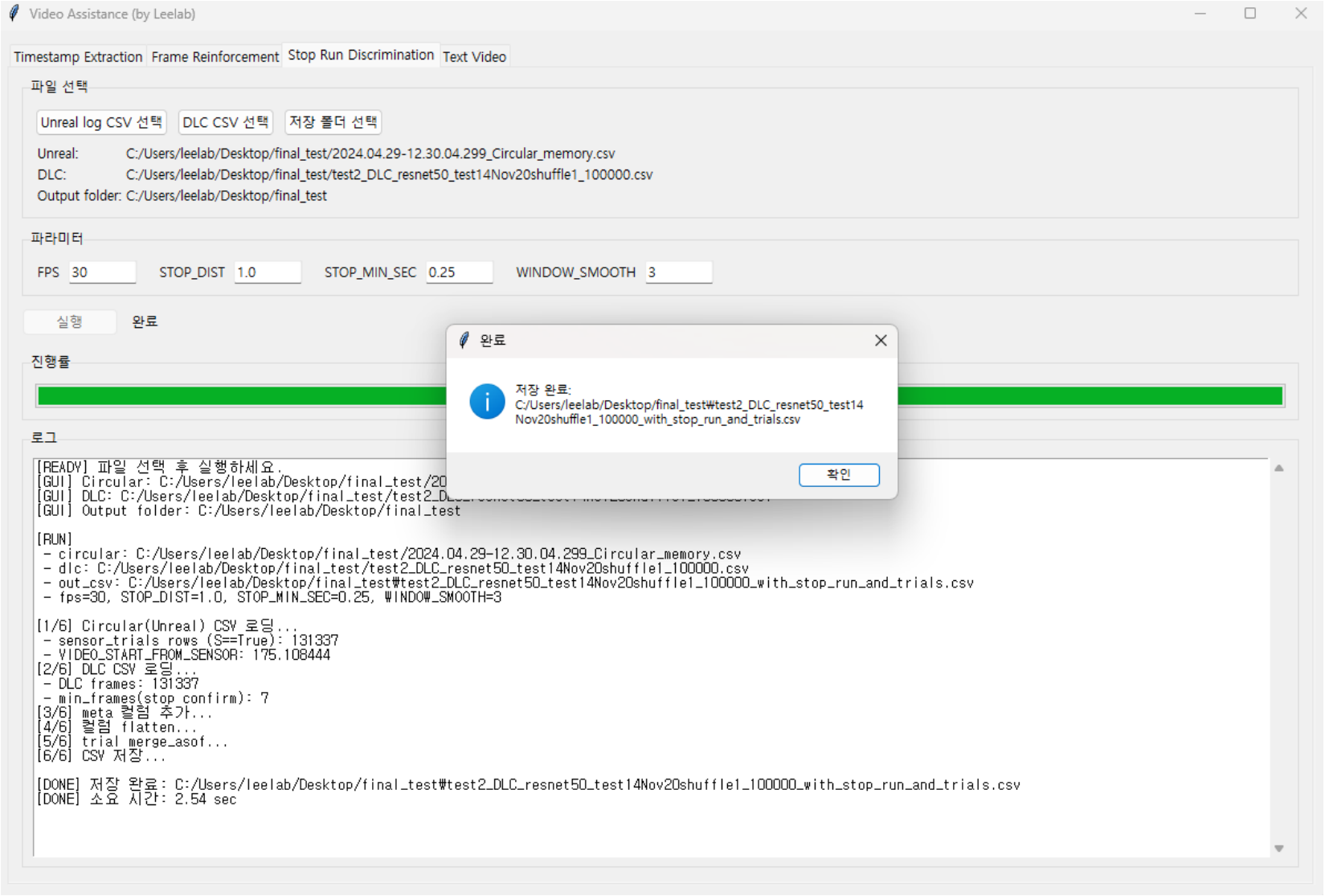
D. Smoothing 변수 값 – smoothing을 얼마나 줄 것인가 (3이 기본 조건)

3. Stop Run Discrimination



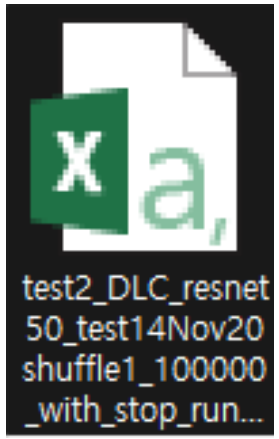
실행

3. Stop Run Discrimination



실행 완료

3. Stop Run Discrimination



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	scorer_boc	DLC_resne	DLC_resne	DLC_resne	DLC_resne	DLC_resne	DLC_resne	meta_leg_	meta_leg_	meta_dist	meta_dist_	meta_stop	meta_fram	trial_F	trial_G
2	0	580.5129	322.6625	0.854215	578.6444	323.3225	0.87977	579.5787	322.9925	0	0	1	0	1	1
3	1	482.0904	300.2362	0.998382	493.0425	295.6533	0.010099	487.5664	297.9448	95.36059	47.6803	1	1	1	1
4	2	385.2881	326.6183	0.974744	487.0127	325.5013	0.151552	436.1504	326.0598	58.60088	51.32049	1	2	1	1
5	3	473.9662	345.544	0.999179	476.1042	340.2759	0.013264	475.0352	342.9099	42.37877	65.44675	1	3	1	1
6	4	608.0491	322.0126	0.989371	619.1069	318.4738	0.044213	613.578	320.2432	140.3847	80.4548	1	4	1	1
7	5	499.4873	298.8803	0.989501	721.2505	118.8361	0.014961	610.3689	208.8582	111.4312	98.06489	1	5	1	1
8	6	415.8175	315.5934	0.926241	417.9351	311.8076	0.94852	416.8763	313.7005	220.0712	157.2957	1	6	1	1
9	7	456.3912	341.0372	0.107016	458.2773	338.6851	0.96814	457.3343	339.8611	48.1791	126.5605	1	7	1	1
10	8	561.1155	325.9793	0.957214	563.1508	324.2196	0.701022	562.1332	325.0994	105.8335	124.6946	1	8	1	1
11	9	494.8902	294.257	0.99909	719.1444	120.099	0.007562	607.0173	207.178	126.1747	93.39574	1	9	1	1
12	10	393.5537	317.7222	0.438722	501.605	318.6631	0.4843	447.5793	318.1926	194.28	142.096	1	10	1	1
13	11	473.399	345.2799	0.998295	479.6003	341.554	0.02046	476.4996	343.417	38.37517	119.6099	1	11	1	1
14	12	601.7471	326.5215	0.983877	607.8585	324.1533	0.028957	604.8028	325.3374	129.5707	120.7419	1	12	1	1
15	13	508.1038	303.5594	0.990144	514.5222	300.2678	0.041966	511.313	301.9136	96.37958	88.10849	1	13	1	1
16	14	418.0007	312.5953	0.734158	418.3831	311.6598	0.989552	418.1919	312.1275	93.67954	106.5433	1	14	1	1
17	15	449.9694	343.4333	0.958487	451.8745	341.2004	0.034408	450.922	342.3168	44.52699	78.19537	1	15	1	1
18	16	553.4557	328.569	0.976488	559.1426	326.8525	0.495223	556.2992	327.7108	106.3846	81.53038	1	16	1	1
19	17	501.3015	294.9591	0.999394	510.4589	293.3834	0.011949	505.8802	294.1712	60.55552	70.48905	1	17	1	1
20	18	396.229	318.7567	0.071565	505.6002	319.8571	0.262194	450.9146	319.3069	60.44018	75.79344	1	18	1	1
21	19	464.2526	348.5561	0.99617	467.7665	344.4148	0.033236	466.0096	346.4854	31.08909	50.69493	1	19	1	1
22	20	610.5837	327.3042	0.998547	616.8962	324.3534	0.053142	613.74	325.8288	149.1676	80.23229	1	20	1	1
23	21	500.252	301.8459	0.981636	919.8559	163.7018	0.01625	710.0539	232.7738	133.9239	104.7268	1	21	1	1
24	22	417.7508	320.7252	0.708283	420.7134	317.1664	0.975164	419.2321	318.9458	303.3199	195.4704	1	22	1	1
25	23	462.3677	343.1767	0.953461	465.7653	340.9812	0.299125	464.0665	342.0789	50.45059	162.5648	1	23	1	1
26	24	586.6305	326.0499	0.99513	582.14	325.0561	0.048752	584.3853	325.553	121.4484	158.4063	1	24	1	1
27	25	457.5064	288.9078	0.834544	552.9518	290.2225	0.041966	505.2291	289.5652	86.95299	86.284	1	25	1	1
28	26	476.3108	333.555	0.01834	530.8306	321.8817	0.04792	503.5707	327.7183	38.18918	82.19686	1	26	1	1
29	27	500.7687	342.7258	0.998719	720.9901	120.963	0.056864	610.8794	231.8444	143.8992	89.68045	1	27	1	1
30	28	632.9266	320.9326	0.368133	648.1593	323.7428	0.003654	640.543	322.3377	95.23111	92.43983	1	28	1	1
31	29	474.615	299.0818	0.063244	469.6044	304.3842	0.288576	472.1097	301.733	169.6889	136.2731	1	29	1	1
32	30	418.9065	318.7983	0.965465	419.4293	317.03	0.033066	419.1679	317.9142	55.35939	106.7598	1	30	1	1

stop/run 판단 data 출력

3. Stop Run Discrimination

meta_leg_center_x or y : 중앙 x,y좌표

meta_dist : 평균 중앙 좌표

meta_dist_smooth : smoothing 된 평균 중앙 좌표

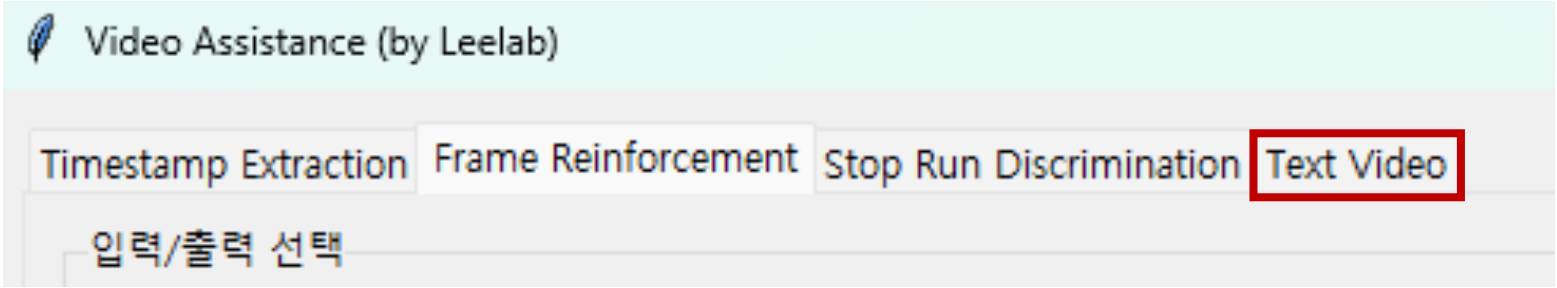
meta_stop : 1 – run 0- stop

trial_F : 현재 trial

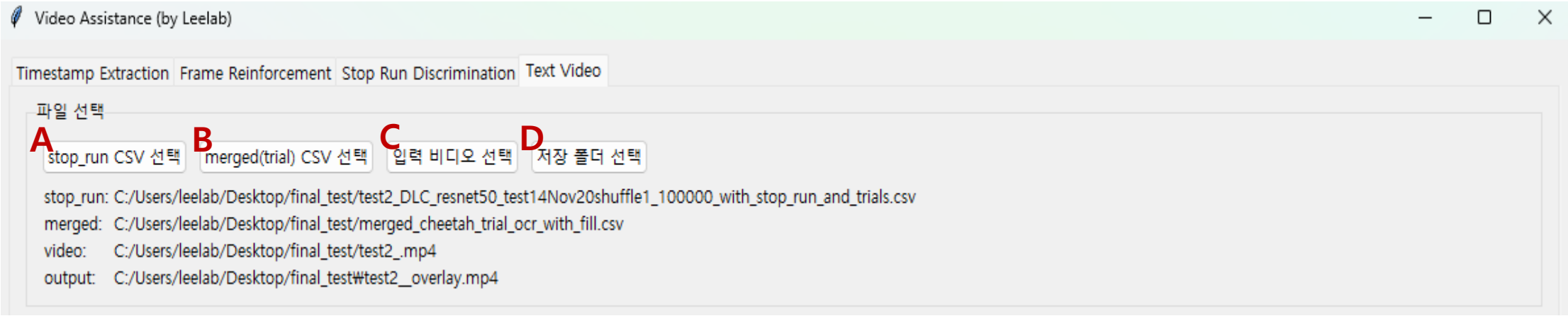
trial_G : 누적 trial

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	scorer_boc	DLC_resne	DLC_resne	DLC_resne	DLC_resne	DLC_resne	DLC_resne	meta_leg_	meta_leg_	meta_dist	meta_dist_	meta_stop	meta_fram	trial_F	trial_G
2	0	580.5129	322.6625	0.854215	578.6444	323.3225	0.87977	579.5787	322.9925	0	0	1	0	1	1
3	1	482.0904	300.2362	0.998382	493.0425	295.6533	0.010099	487.5664	297.9448	95.36059	47.6803	1	1	1	1
4	2	385.2881	326.6183	0.974744	487.0127	325.5013	0.151552	436.1504	326.0598	58.60088	51.32049	1	2	1	1
5	3	473.9662	345.544	0.999179	476.1042	340.2759	0.013264	475.0352	342.9099	42.37877	65.44675	1	3	1	1
6	4	608.0491	322.0126	0.989371	619.1069	318.4738	0.044213	613.578	320.2432	140.3847	80.4548	1	4	1	1
7	5	499.4873	298.8803	0.989501	721.2505	118.8361	0.014961	610.3689	208.8582	111.4312	98.06489	1	5	1	1
8	6	415.8175	315.5934	0.926241	417.9351	311.8076	0.94852	416.8763	313.7005	220.0712	157.2957	1	6	1	1
9	7	456.3912	341.0372	0.107016	458.2773	338.6851	0.96814	457.3343	339.8611	48.1791	126.5605	1	7	1	1
10	8	561.1155	325.9793	0.957214	563.1508	324.2196	0.701022	562.1332	325.0994	105.8335	124.6946	1	8	1	1
11	9	494.8902	294.257	0.99909	719.1444	120.099	0.007562	607.0173	207.178	126.1747	93.39574	1	9	1	1
12	10	393.5537	317.7222	0.438722	501.605	318.6631	0.4843	447.5793	318.1926	194.28	142.096	1	10	1	1
13	11	473.399	345.2799	0.998295	479.6003	341.554	0.02046	476.4996	343.417	38.37517	119.6099	1	11	1	1
14	12	601.7471	326.5215	0.983877	607.8585	324.1533	0.028957	604.8028	325.3374	129.5707	120.7419	1	12	1	1
15	13	508.1038	303.5594	0.990144	514.5222	300.2678	0.041966	511.313	301.9136	96.37958	88.10849	1	13	1	1
16	14	418.0007	312.5953	0.734158	418.3831	311.6598	0.989552	418.1919	312.1275	93.67954	106.5433	1	14	1	1
17	15	449.9694	343.4333	0.958487	451.8745	341.2004	0.034408	450.922	342.3168	44.52699	78.19537	1	15	1	1
18	16	553.4557	328.569	0.976488	559.1426	326.8525	0.495223	556.2992	327.7108	106.3846	81.53038	1	16	1	1
19	17	501.3015	294.9591	0.999394	510.4589	293.3834	0.011949	505.8802	294.1712	60.55552	70.48905	1	17	1	1
20	18	396.229	318.7567	0.071565	505.6002	319.8571	0.262194	450.9146	319.3069	60.44018	75.79344	1	18	1	1
21	19	464.2526	348.5561	0.99617	467.7665	344.4148	0.033236	466.0096	346.4854	31.08909	50.69493	1	19	1	1
22	20	610.5837	327.3042	0.998547	616.8962	324.3534	0.053142	613.74	325.8288	149.1676	80.23229	1	20	1	1
23	21	500.252	301.8459	0.981636	919.8559	163.7018	0.01625	710.0539	232.7738	133.9239	104.7268	1	21	1	1
24	22	417.7508	320.7252	0.708283	420.7134	317.1664	0.975164	419.2321	318.9458	303.3199	195.4704	1	22	1	1
25	23	462.3677	343.1767	0.953461	465.7653	340.9812	0.299125	464.0665	342.0789	50.45059	162.5648	1	23	1	1
26	24	586.6305	326.0499	0.99513	582.14	325.0561	0.048752	584.3853	325.553	121.4484	158.4063	1	24	1	1
27	25	457.5064	288.9078	0.834544	552.9518	290.2225	0.041966	505.2291	289.5652	86.95299	86.284	1	25	1	1
28	26	476.3108	333.555	0.01834	530.8306	321.8817	0.04792	503.5707	327.7183	38.18918	82.19686	1	26	1	1
29	27	500.7687	342.7258	0.998719	720.9901	120.963	0.056864	610.8794	231.8444	143.8992	89.68045	1	27	1	1
30	28	632.9266	320.9326	0.368133	648.1593	323.7428	0.003654	640.543	322.3377	95.23111	92.43983	1	28	1	1
31	29	474.615	299.0818	0.063244	469.6044	304.3842	0.288576	472.1097	301.733	169.6889	136.2731	1	29	1	1
32	30	418.9065	318.7983	0.965465	419.4293	317.03	0.033066	419.1679	317.9142	55.35939	106.7598	1	30	1	1

4. Text Video


Video Assistance (by Leelab)
Timestamp Extraction
Frame Reinforcement
Stop Run Discrimination
Text Video
입력/출력 선택

4. Text Video



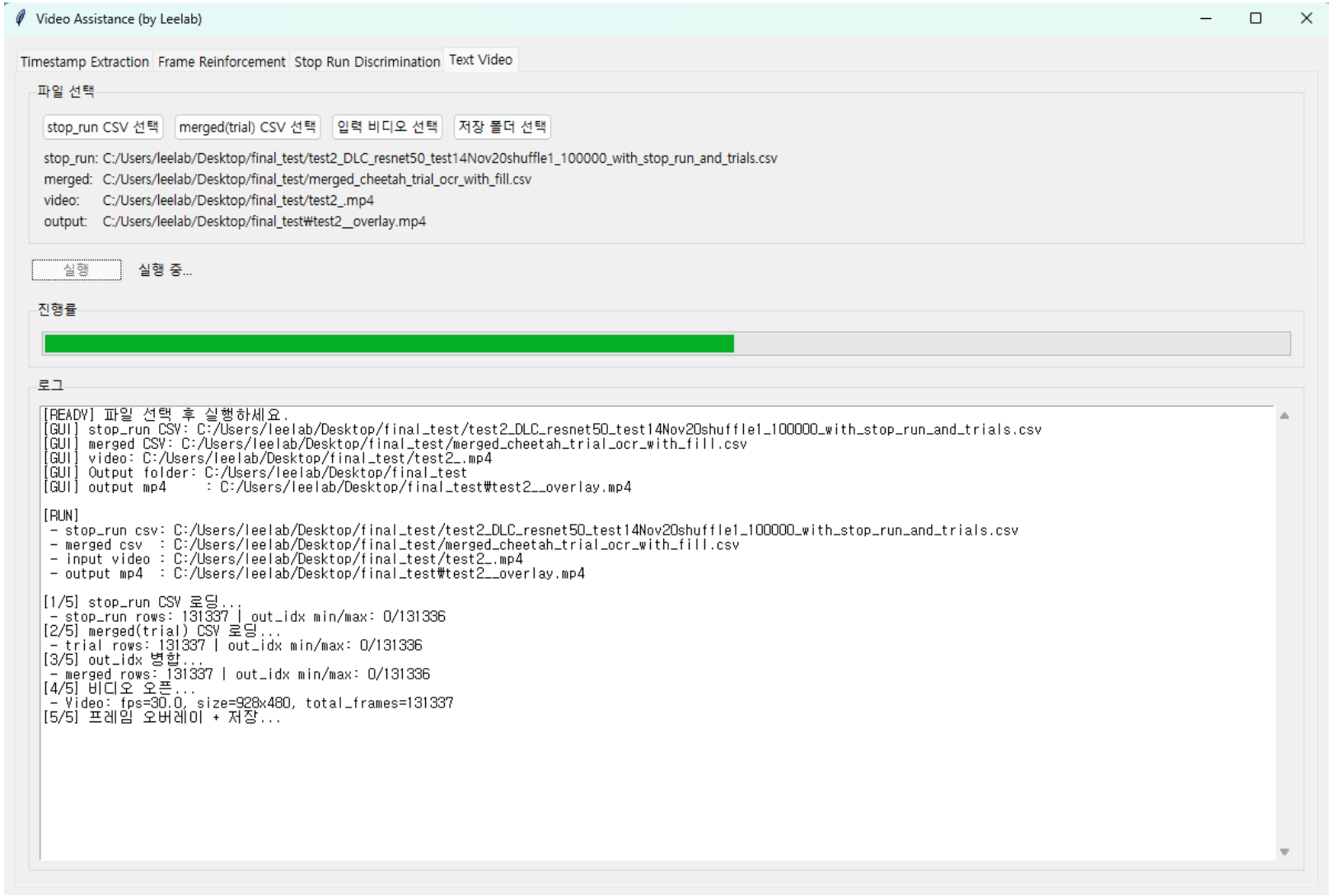
A. Stop Run Discrimination 에서 출력된 stop/run csv input

B. Timestamp Extraction 에서 출력된 align csv input

C. Text를 입힐 input 영상 선택 – Frame Reinforcement에서 frame 보강 된 영상

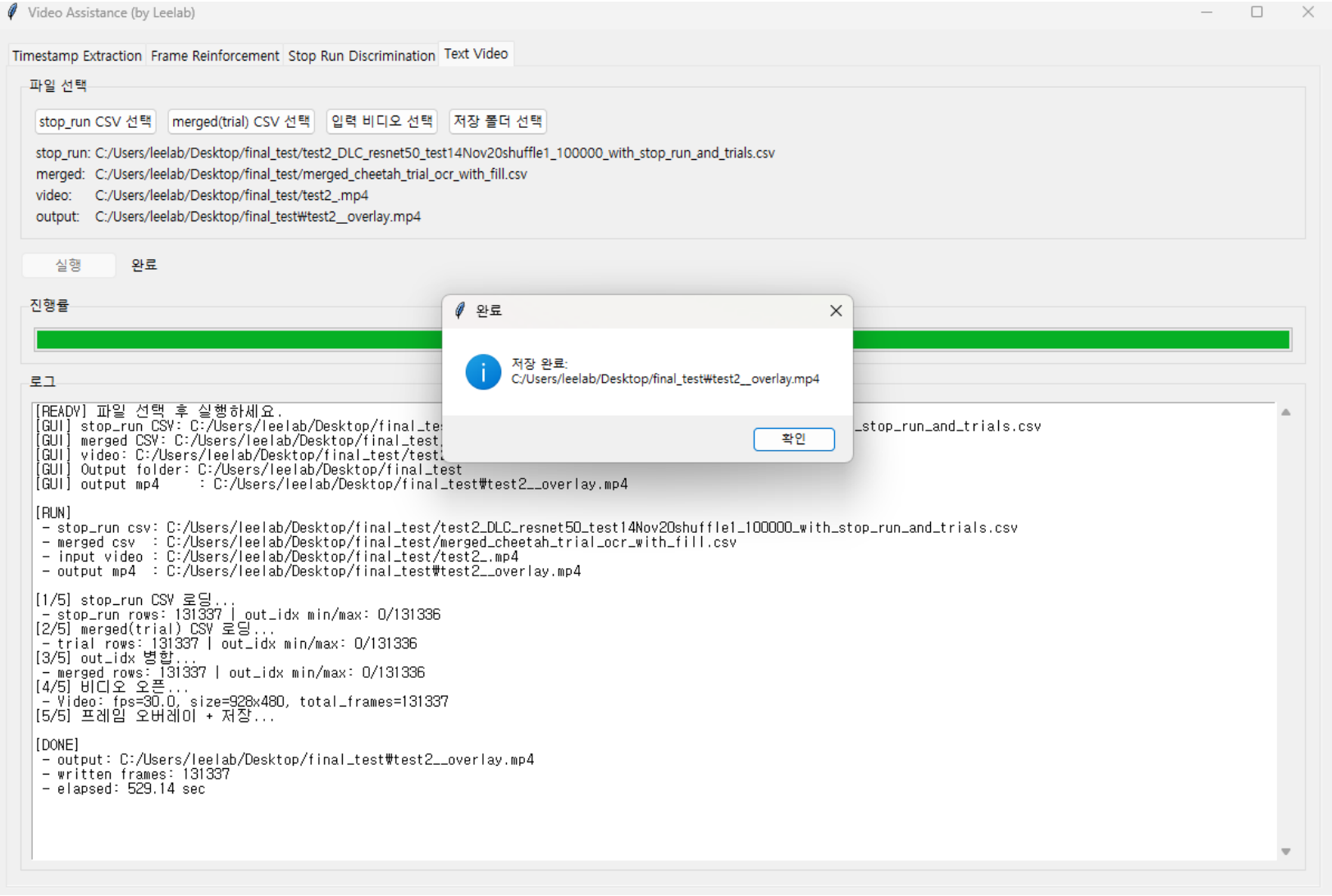
D. 저장할 경로 선택

4. Text Video



실행

4. Text Video



실행 완료

4. Text Video

출력된 영상